

AI STUDY ASSISTANT

Python • NLP • Generative AI

TECHNICAL PROJECT PRESENTATION

Concept / planned implementation unless code is available. Prepared for portfolio / recruiter discussion.

Project Overview

Turning study material into an interactive experience



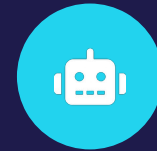
What it does

An AI study assistant designed to turn educational documents into an interactive learning experience.



What it enables

Planned capabilities include document Q&A, quizzes and flashcards, generated directly from the source material.



AI STUDY ASSISTANT

Ask questions.
Generate quizzes.
Build flashcards.

Problem Statement

Why studying from static material falls short



Slow to search

Students often spend time hunting through long PDFs for one specific piece of information.



Static notes

Notes don't automatically adapt to what a learner actually wants to revise right now.

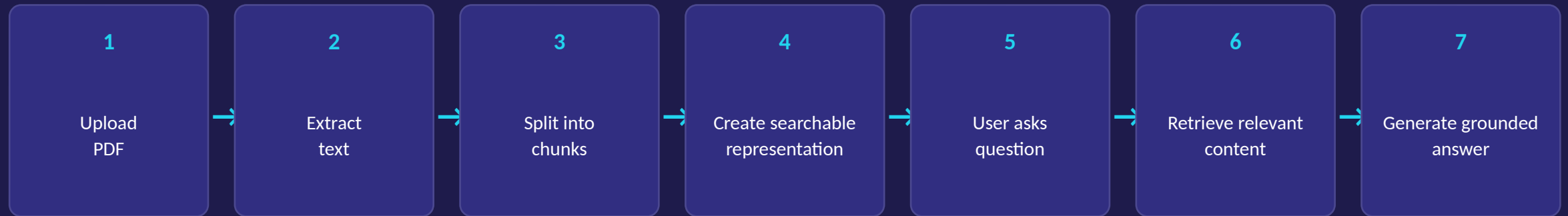


The opportunity

An AI assistant can make study material searchable, interactive, and responsive to questions.

Document Q&A Flow

From a raw PDF to a grounded answer



Why retrieval matters

Grounding the answer in retrieved content keeps it tied to the actual study material.

Planned Features

Core capabilities, with room for more



PDF / document question answering



Quiz generation



Flashcard generation



Topic-based revision

POTENTIAL ADDITIONS

- Conversation history
- Progress tracking

Technology Direction

The stack this concept is built around



Python

Orchestration and processing.



PDF extraction library

Document ingestion.



NLP / LLM layer

Question answering and generation.



Vector database (optional)

Embeddings for semantic retrieval.



Web interface

The layer for interaction.

Example

A walkthrough of one question, end to end



Input document

Lecture notes on machine learning.



Question

"What is overfitting?"



Retrieval + answer

The system retrieves the relevant section and produces an answer grounded in that material.



Quiz mode

Can generate practice questions from that same source material — turning the note into revision, not just an answer.

Important Design Consideration

Keeping answers trustworthy



Grounded answers

Answers should be grounded in the supplied study material wherever possible.



Retrieval reduces noise

Retrieval helps reduce irrelevant or unrelated responses.



Clear distinction

The system should clearly separate document-supported answers from general knowledge.

Future Improvements

Where the assistant could go next



Citation of source passages



Adaptive difficulty



Personalized revision schedule



Voice interaction



Progress analytics



Deployment with auth & cloud storage